

The background features several abstract, overlapping geometric shapes in various shades of blue. These shapes are primarily located on the left side and top corners, creating a modern, minimalist aesthetic. The shapes include triangles, trapezoids, and rounded polygons, some with sharp edges and others with smooth, curved borders.

# **Agents Architectures**

**Week 6**

# Last Week

## Week 4: Tools for building AI Agents

LangChain, Azure Open AI Api, LLMMod.ai

## Week 5: AI Workflow & ReAct Agent

AI Workflow, ReAct Agent, Individual Project Launch

# HR departments are overwhelmed

## Problem Statement

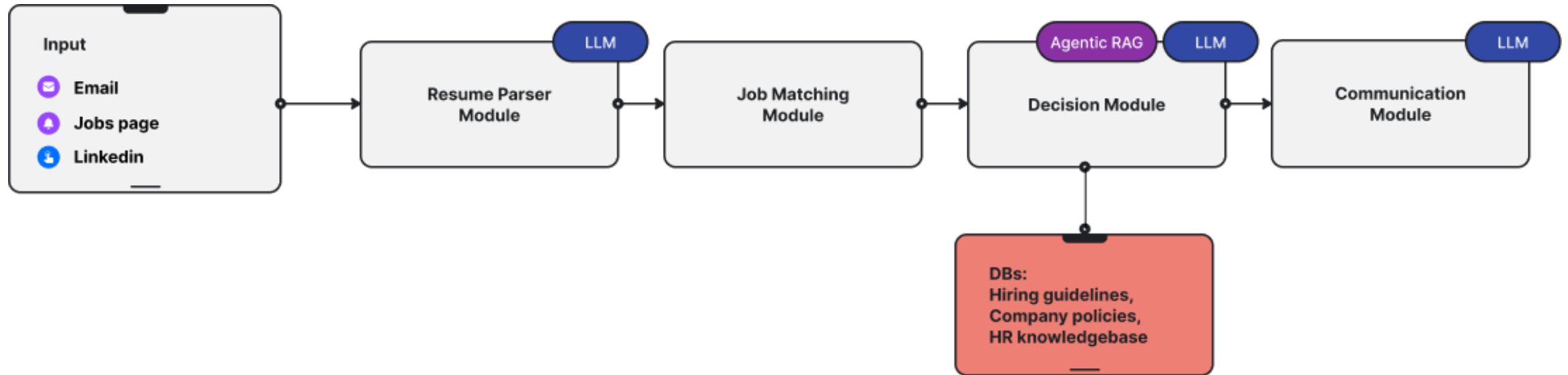
📁 HR departments receive a high volume of resumes for job openings, making it challenging to efficiently and fairly evaluate candidates.

🕒 Manual sorting is time-consuming and prone to human biases.

🔄 Communication with candidates is also time consuming and difficult.



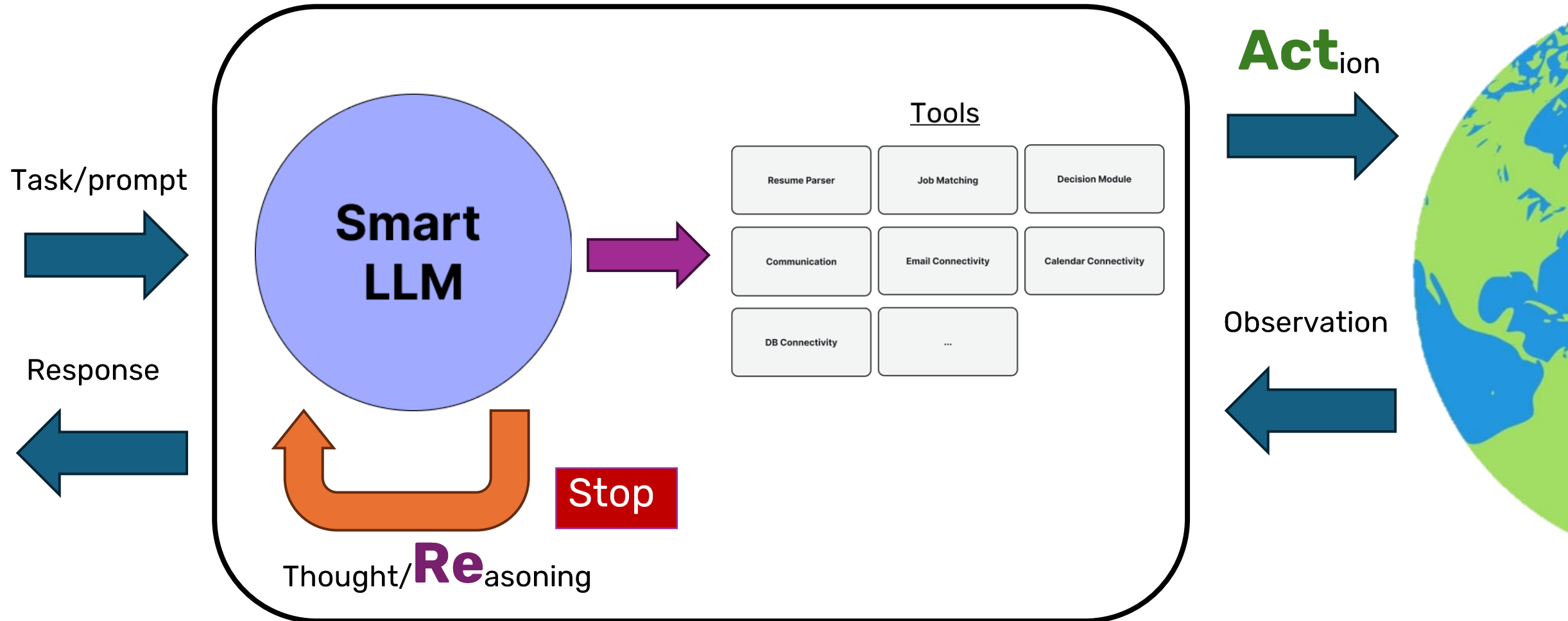
# Toby, Auto HR



# AI Workflow - Pros

- ✓ Simple and reliable
- ✓ Cheap to run
- ✓ Easy to understand and debug
- ✓ Easy to scale
- ✓ Great for some tasks

# ReAct Agent



# This Week

## Week 4: Tools for building AI Agents

LangChain, Azure Open AI Api, LLMMod.ai

## Week 5: AI Workflow & ReAct Agent

AI Workflow, ReAct Agent, Individual Project Launch

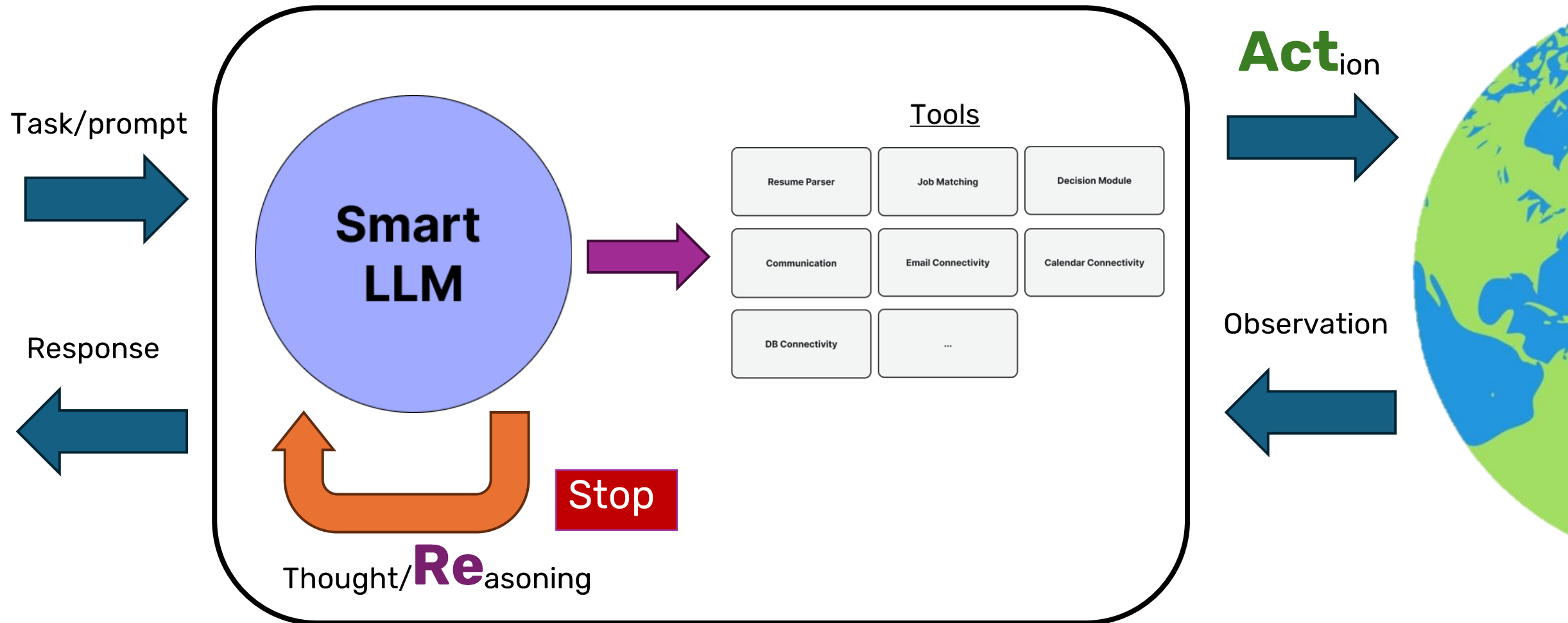
## Week 6: Agents Architectures

Cont. ReAct agent, AI Agents Architectures

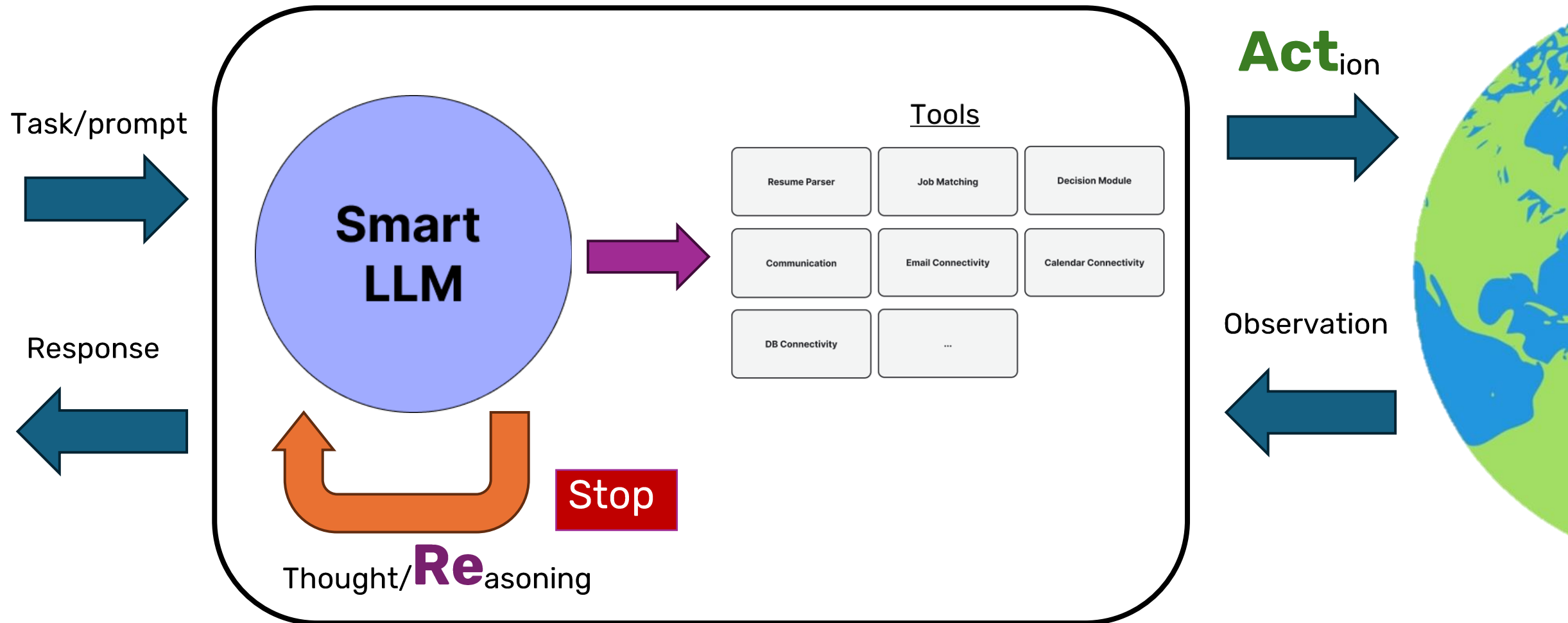
# ReAct Agent Cont.



# ReAct Agent



# ReAct Agent



# ReAct (Reasoning + Acting) Agent

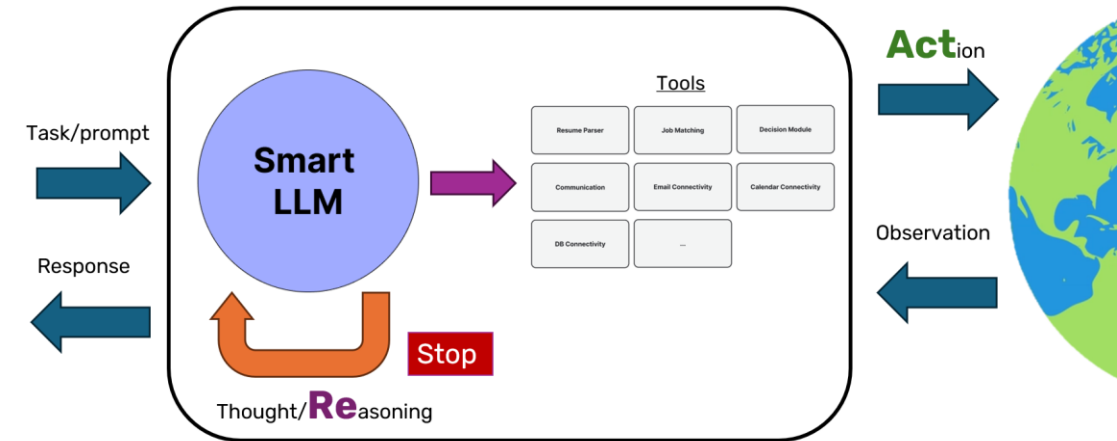
The most widely adopted pattern. The agent iterates through: Thought -> Action -> Observation cycles. It reasons about what to do, executes a tool, observes the result, and repeats until the task is complete.

## Good for

Multi-step problem solving with external tools

Information retrieval and research tasks

Exploratory tasks



# ReAct (Reasoning + Acting) Agent

## Pros

Highly flexible and adaptive

Strong performance in tool-heavy tasks

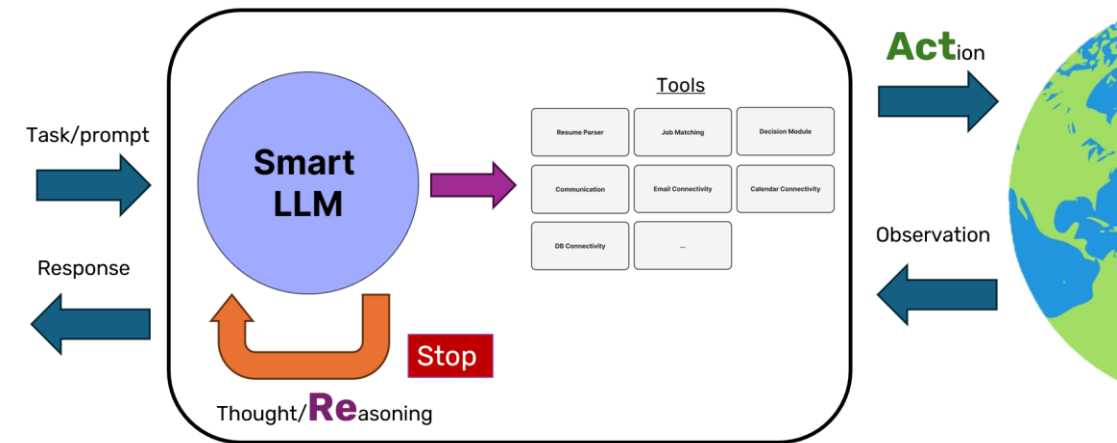
Naturally handles partial or evolving information

## Cons

Harder to audit and debug

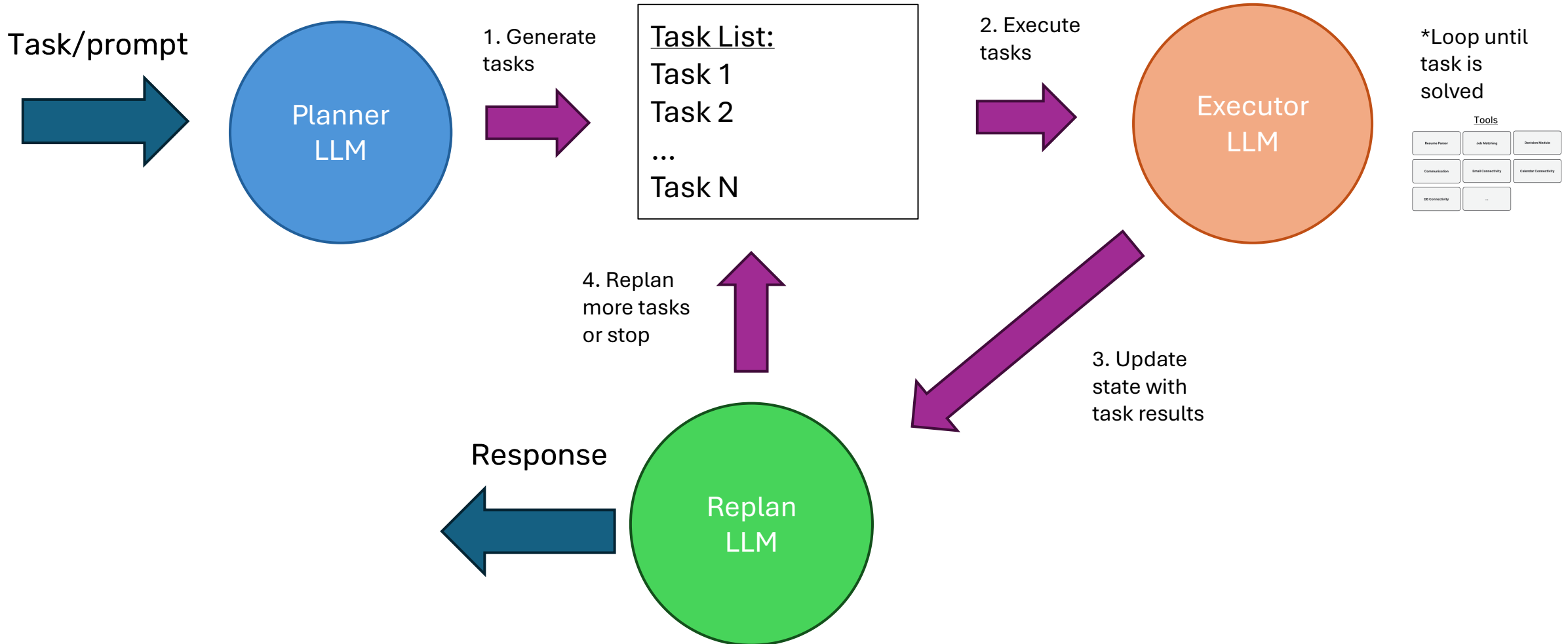
Can enter inefficient loops

Higher risk of reasoning drift without safeguards

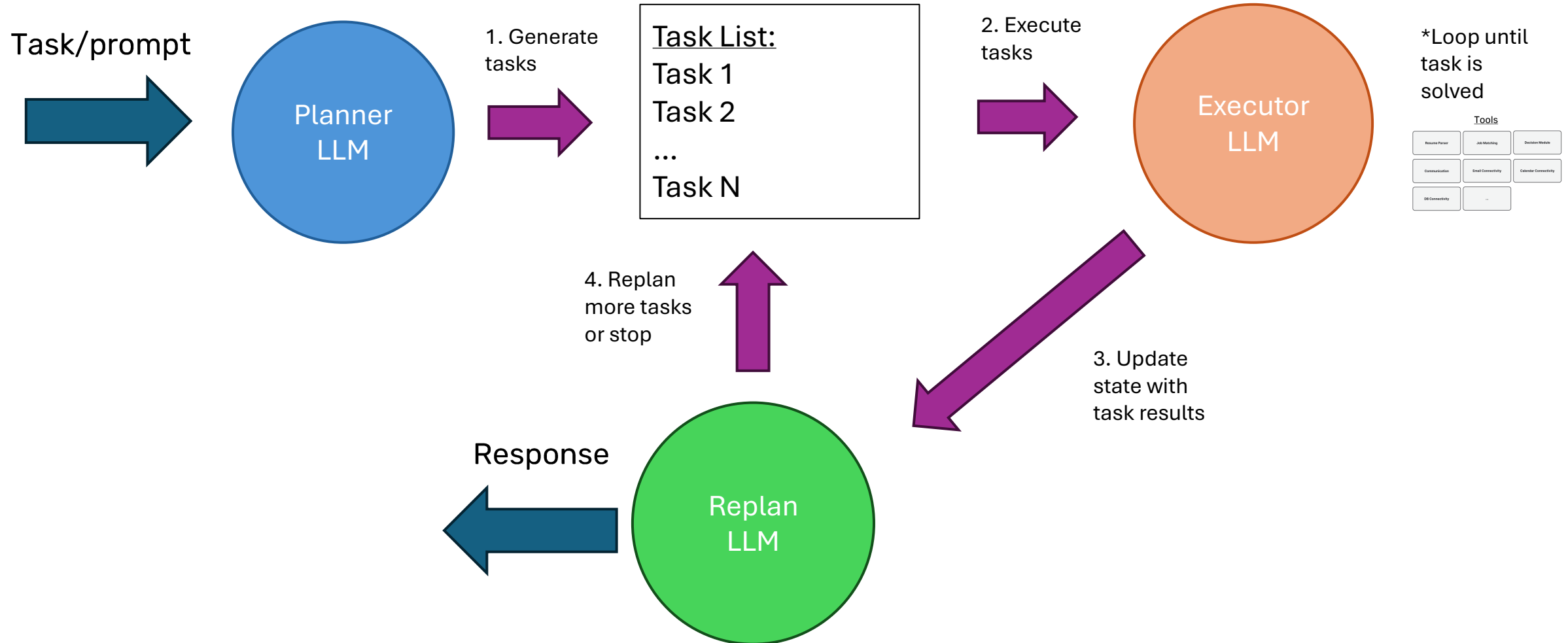


# Plan and Execute Agent

# Plan and Execute Agent



# Plan and Execute Agent



# Plan and Execute Agent

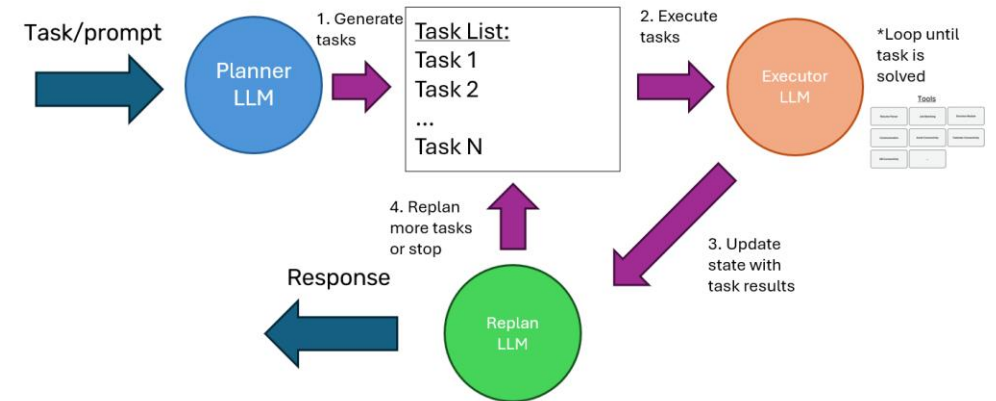
A Plan-and-Execute Agent is an AI system that separates thinking from doing. It first creates a complete plan of steps needed to accomplish a task, then executes each step sequentially—optionally revising the plan as new information emerges.

## Good for

Multi-step workflows with clear structure

Tool orchestration (Search, Code Execution, etc.)

Auditability and human oversight





# Plan and Execute Agent

## Pros

Clear, interpretable execution flow

Easier debugging and monitoring

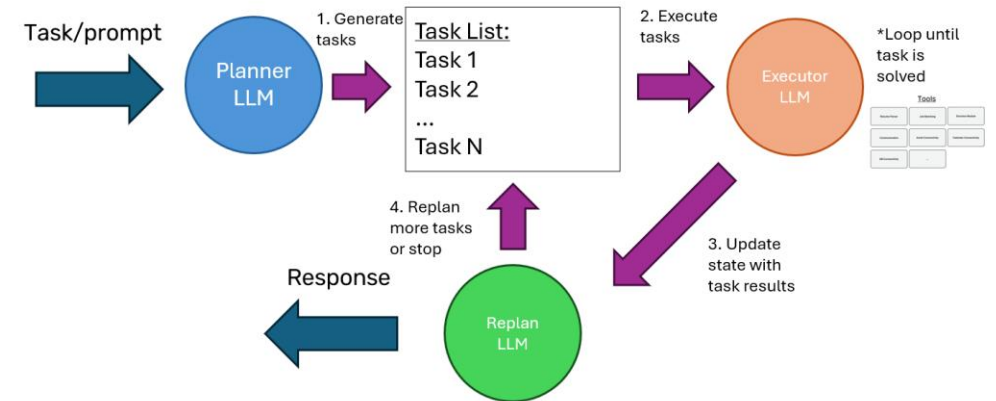
Safer for production workflows

## Cons

Rigid plans may fail in dynamic environments

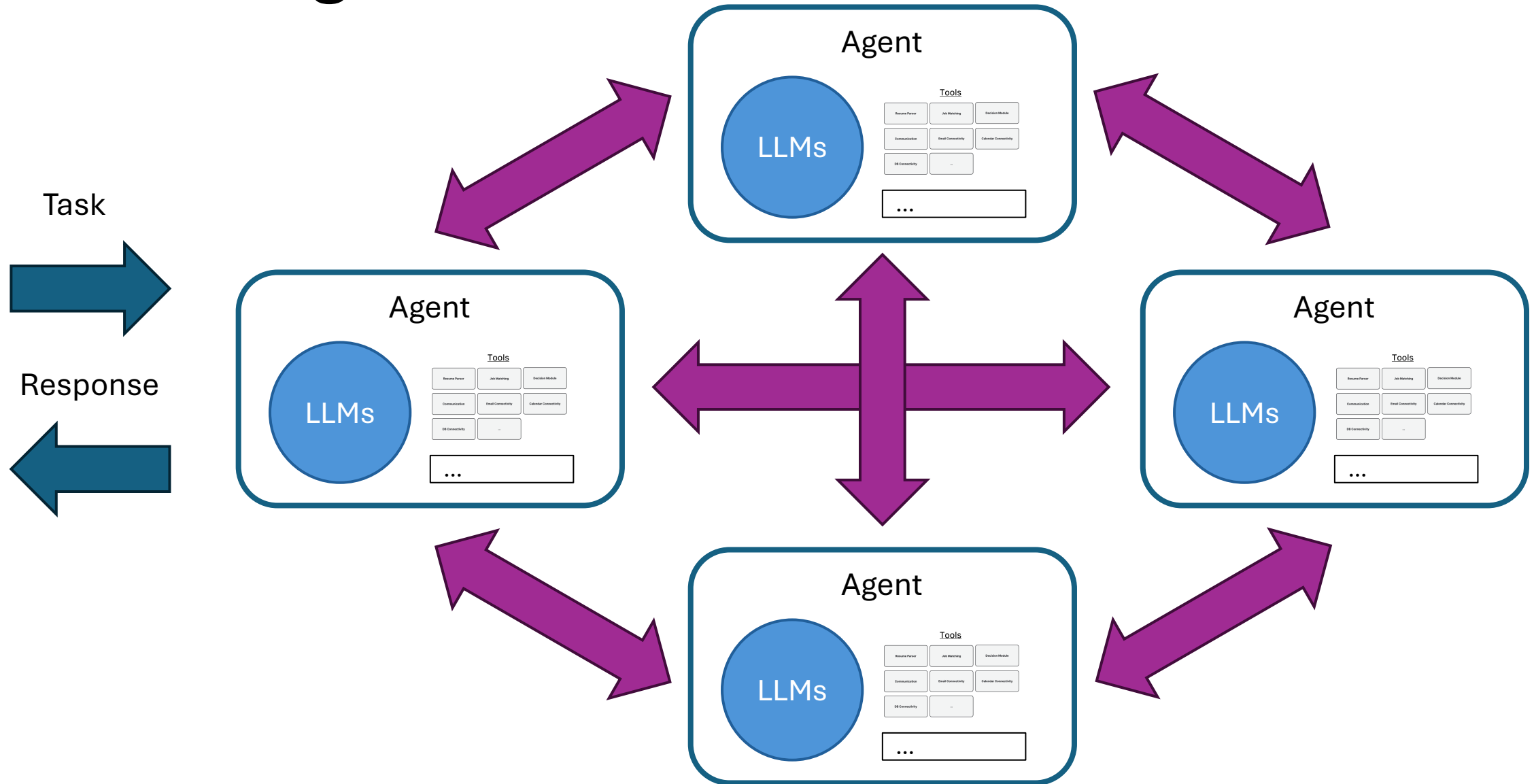
Planning errors propagate to execution

Less flexible than reactive agents



# Multi AI Agents

# Multi AI Agents



# Multi AI Agents

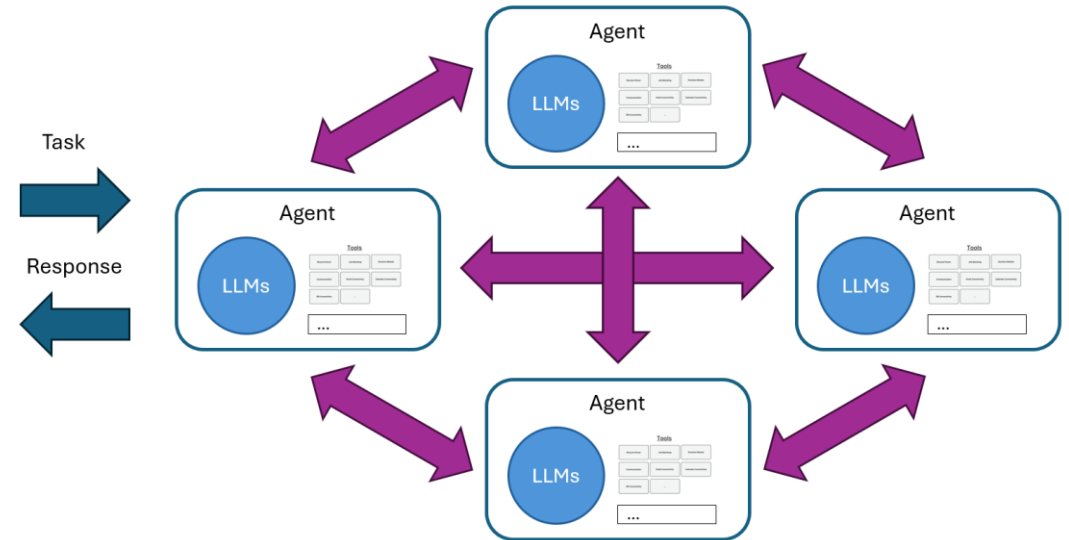
A Multi-Agent system consists of multiple specialized AI agents that collaborate, debate, or divide responsibilities to solve a task collectively.

## Good for

Complex tasks requiring multiple perspectives

Parallel reasoning or decomposition

Simulation, debate, and consensus building



# Multi AI Agents

## Pros

Higher robustness via diversity of reasoning

Parallelism improves coverage and creativity

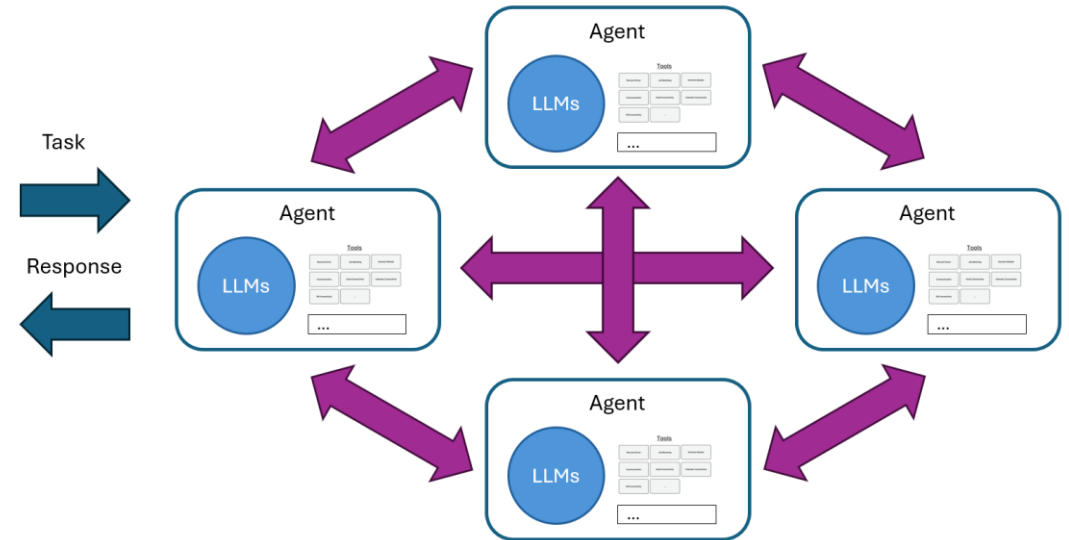
Natural task decomposition

## Cons

Coordination overhead

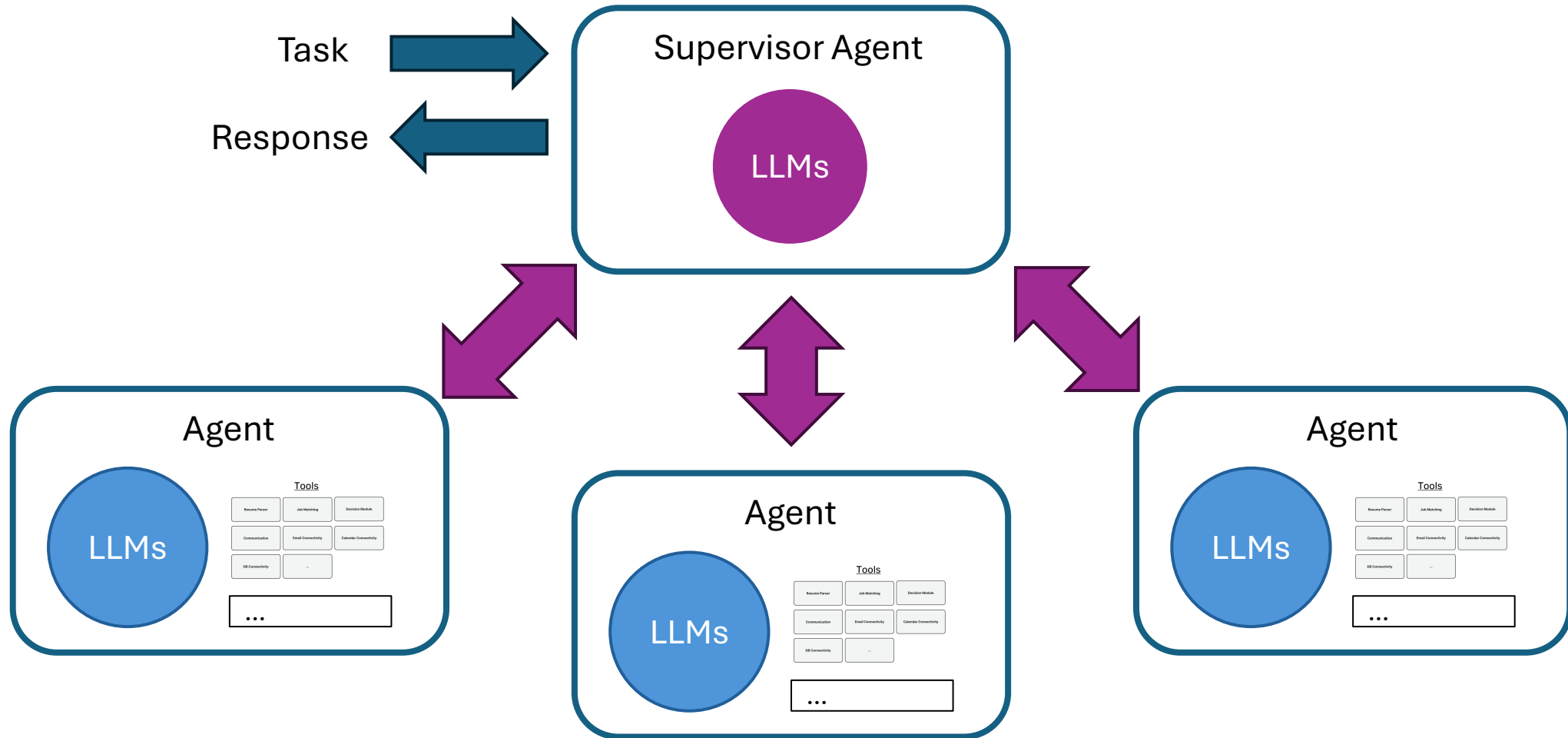
Higher cost and complexity

Risk of conflict or redundant work between agents

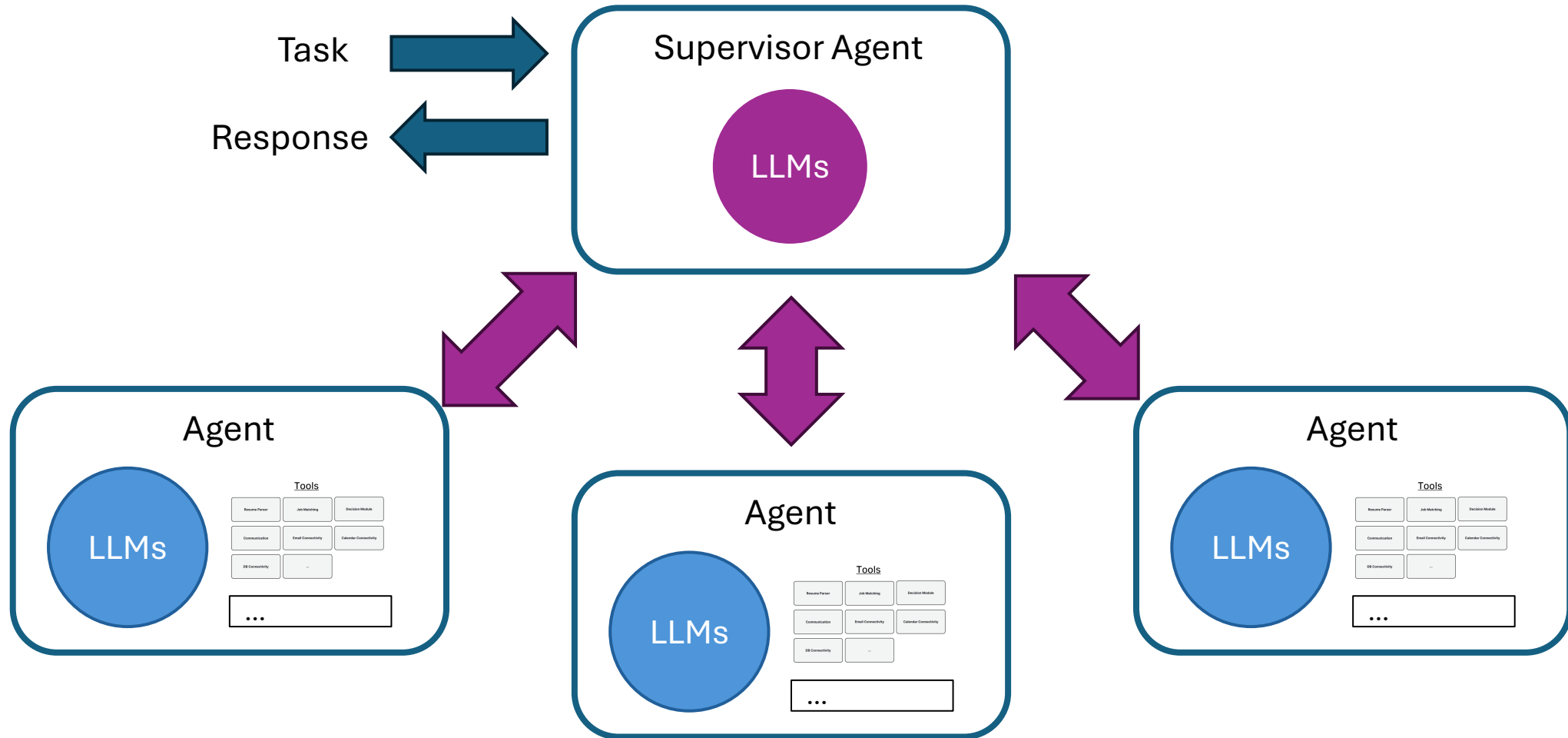


# Supervisor Agent

# Supervisor Agent



# Supervisor Agent





# Supervisor Agent

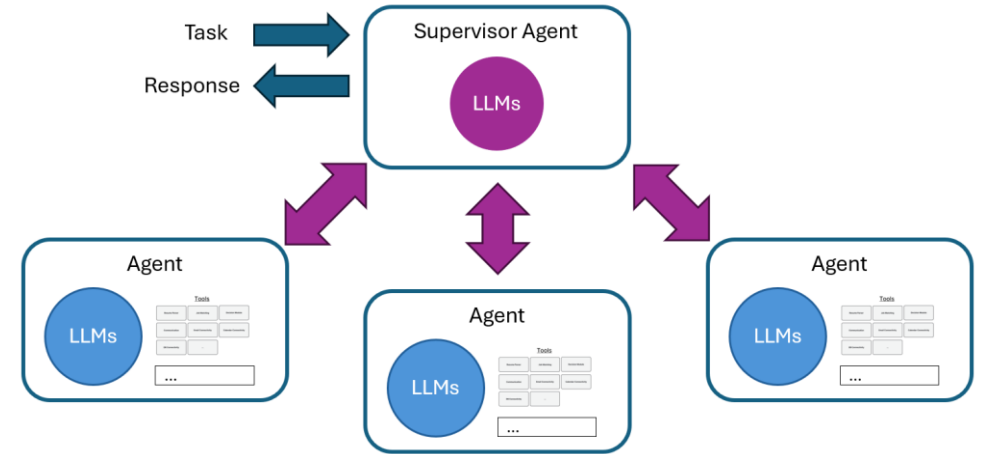
A Supervisor Agent oversees and coordinates other agents. It assigns tasks, monitors progress, resolves conflicts, and decides when to stop or reroute execution.

## Good for

Large agent systems and workflows

Safety-critical or high-stakes tasks

Long-running or multi-stage processes



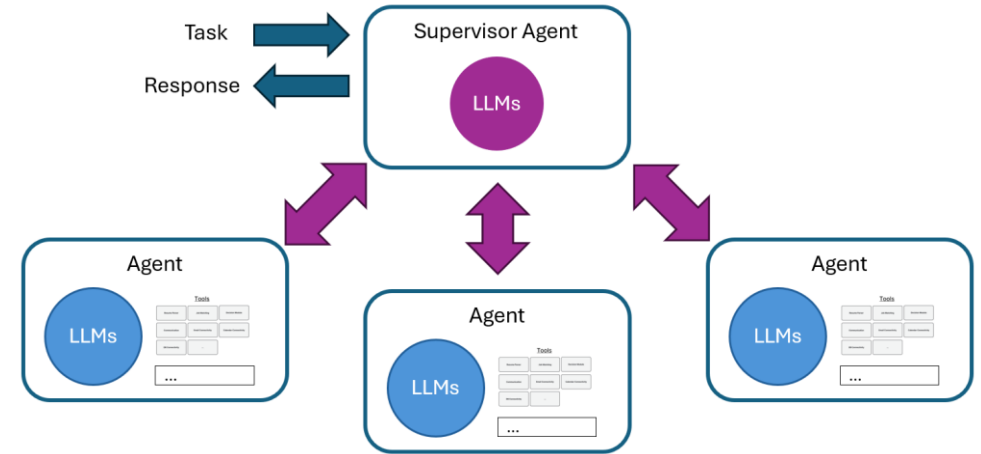
# Supervisor Agent

## Pros

- Centralized control and governance
- Improves reliability and alignment
- Enables scalable agent architectures

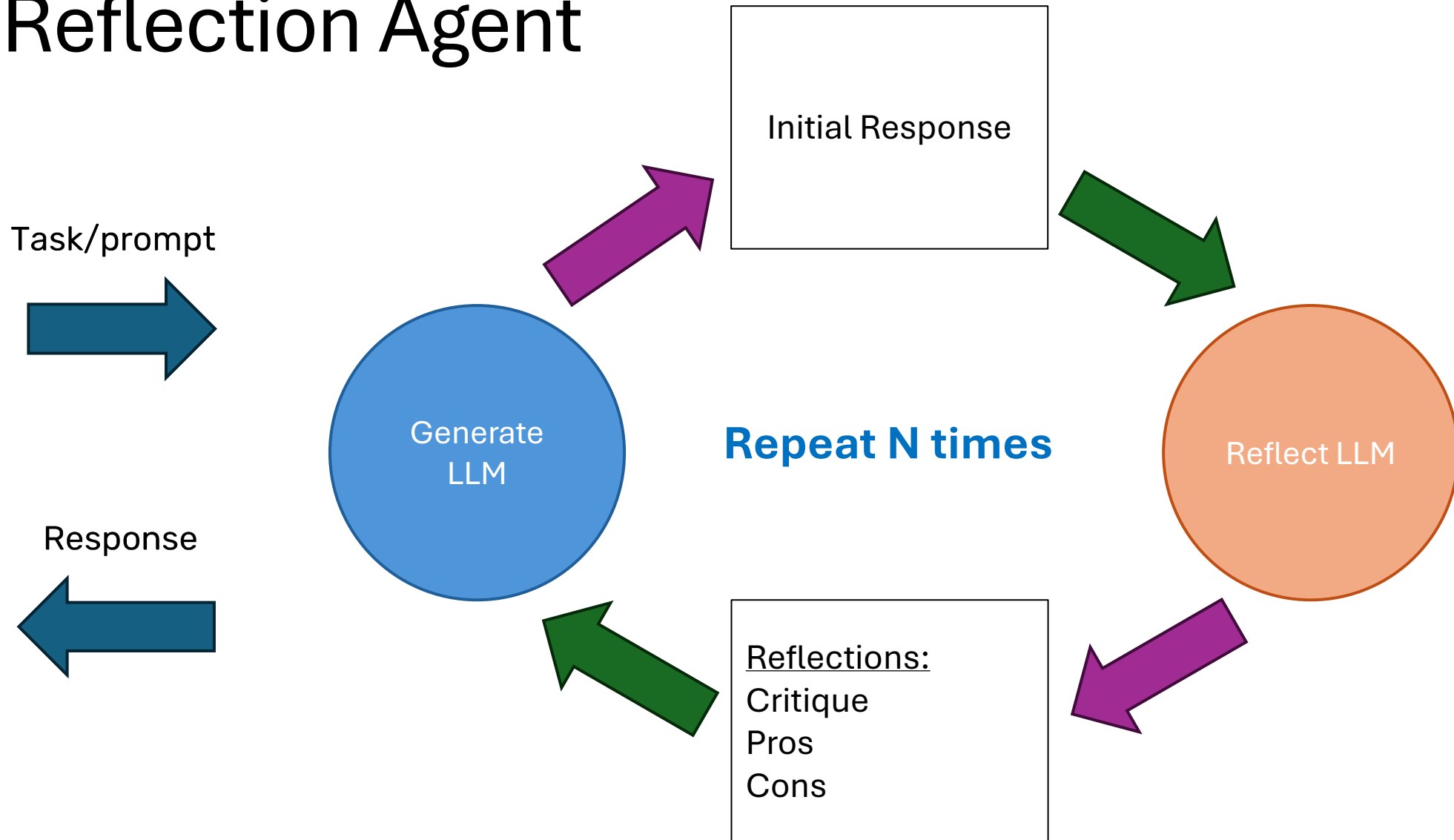
## Cons

- Single point of failure
- Additional latency and design complexity
- Requires careful prompt and policy design

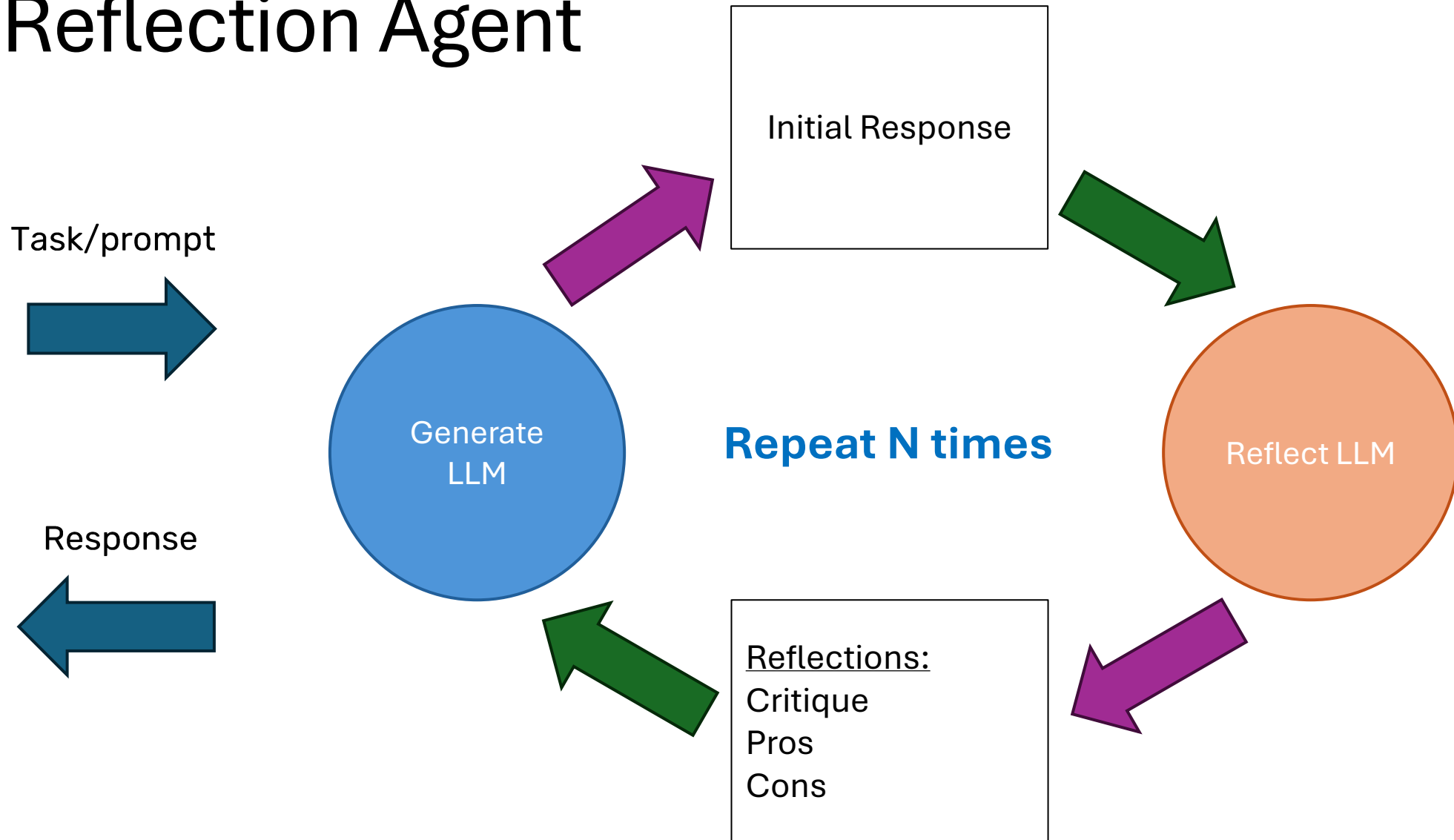


# Reflection Agent

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# Reflection Agent

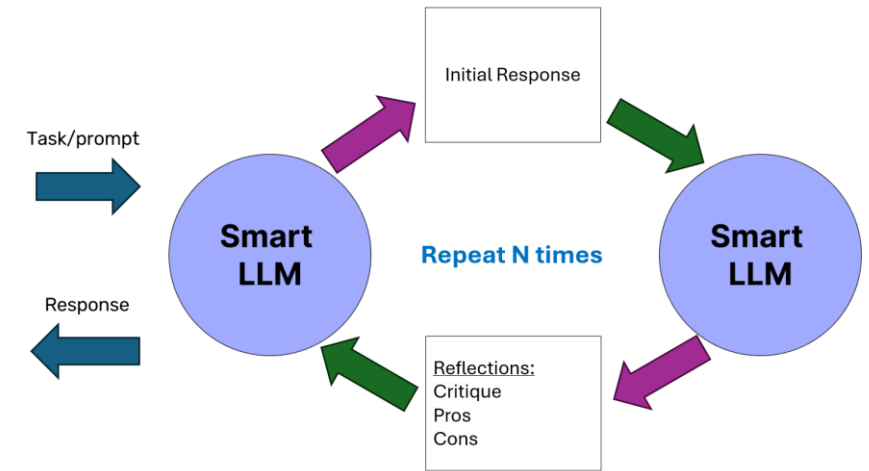
Reflection is a prompting strategy used to improve the quality and success rate of agents and similar AI systems. It involves prompting an LLM to reflect on and critique its past actions, sometimes incorporating additional external information such as tool observations.

## Good for

Complex reasoning tasks (Math problems, logic puzzles)

Writing and content refinement

Decision-making under uncertainty



# Reflection Agent

## Pros

Improves output quality over iterations

Reduces logical errors and hallucinations

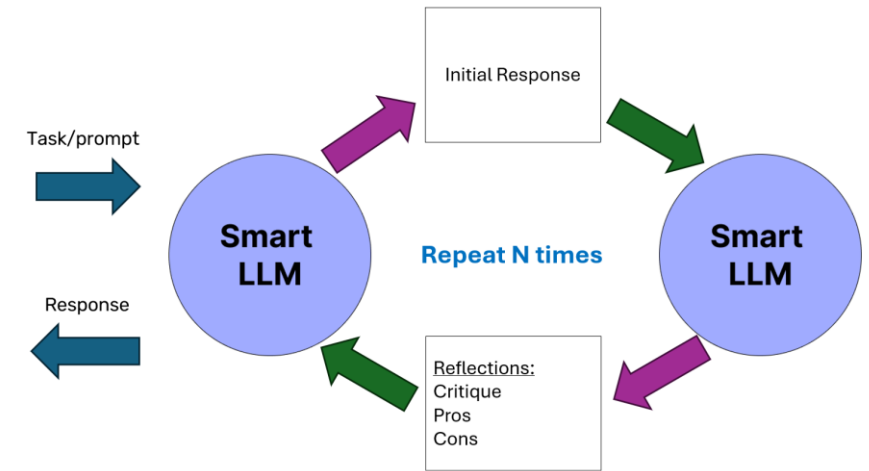
Encourages self-correction and learning from mistakes

## Cons

Higher latency and token cost

Can over-reflect and stall progress

Limited benefit for simple or time-critical tasks



# Project Presentations



# Project presentations

## Dates:

Batch 1: **6.7.2026**

Batch 2: **13.7.2026**

Batch 3: **20.7.2026**

## Bonus:

The best and second-best presentations in each batch will receive **+3 and +2 bonus points**, respectively, added to the final score.

Questions?